

Microeconomics II**Problem set 1****Exercise 1**

Research concerning the electricity market has highlighted the following factors that influence demand: the price of the electricity (P_x), the income of consumers (M), the average level of energy efficiency (E), and the price of gas (P_z). Thus, market demand is:

$$Q_x = f(P_x, M, E, P_z) \quad (1)$$

What is the expected impact of each of the variables identified?

Exercise 2

Graphically analyze the likely impact of the following occurrences on the respective markets.

- (a) Due to the lack of rain in spring, there has been a reduction in the production of cereals.
- (b) A recent study indicates that the consumption of processed meat can be harmful to our health.

Exercise 3

Explain the foreseeable impacts of government price control measures:

- (a) When there is a maximum price that is below the equilibrium price.
- (b) When there is a minimum price that is higher than the equilibrium price.

Exercise 4

Consider the following two goods in an economy: x = Coffee, y = Books. A typical consumer can be described through this utility function

$$U(x, y) = 10x^{0.3}y^{0.3} \quad (2)$$

The prices of goods are: $p_x = 5$ monetary units (m.u.) and $p_y = 10$ m.u. Assume a 400 m.u. income for the consumer.

- (a) Present the optimal consumption plan/choice/bundle.

- (b) Suppose that the consumer has received two books in Christmas. Represent the new budget constraint. Will it be that the new choice is to buy less books (than with the previous budget line) and spend the amount saved to buy more coffee?

Exercise 5

Consider a firm with the following production function:

$$Q(L, K) = 2L^{0.5}K^{0.5} \quad (3)$$

The price per unit of labour is 9 m.u. ($w = 9$) and the price per unit of capital is €9 ($r = 4$).

- (a) Calculate the minimum cost if the firm wants to produce 100 units of the product Q ($Q = 100$).
- (b) Suppose the firm has a budget of 504 m.u. ($C = 504$). With this information, find L and K that maximize output.
- (c) The firm was planning to produce according to the conditions in (b). Suppose it has the opportunity of ensuring financing to produce the 100 units, but these 100 units has to be delivered in the short-run. Find L and K that make this possible.
- (d) Now, suppose that the price of capital is 5 m.u. Find L and K if the firm wants to keep the level of production (Q) as found in (b).